財政學 HW5

1. (Hyman 5.2 改編)

The marginal benefit per gardener varies for five residents according to the following table:

Marginal Benefit per each gardener						
VOTER	1	2	3	4	5	6
Mike	350	300	200	150	100	50
Jan	100	80	60	40	20	0
Franklin	175	150	125	100	90	80
Susan	70	50	30	20	10	0
Megan	120	110	100	90	85	70

Marginal cost of hiring each gardener is $400.

(a) If each voter is assigned a tax share of $80, what is the political equilibrium under majority rule?

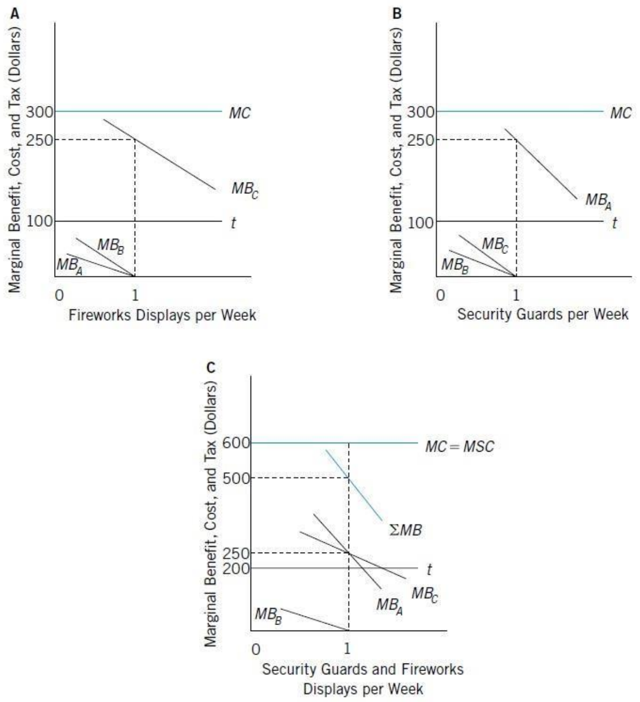
(b) Show that this equilibrium does not result in the efficient output of gardening services.

(c) Show how the Lindahl equilibrium will differ from the political equilibrium under majority rule.

2. (Hyman 5.3 改編)

The example of logrolling used in the text (Figure 5.8) assumes that the transactions costs of vote trading are zero. Suppose instead that voters A and C have to incur expenditures equal to $80 per week to reach agreement on the vote-trading scheme. Show how this would prevent successful logrolling. Also show how logrolling would be impossible if the marginal benefit of the first security guard were only $180 to voter A and transactions costs were zero.

FIGURE 5.8 Logrolling



Logrolling can result in the passage of two issues together that could not pass if voted on separately.

3. (Rosen 6.9)

Assume that the demand curve for milk is given by Q = 100-10P, where P is the price per gallon and Q is the quantity demanded per year. The supply curve is horizontal at the price of 2.

(a) Assuming that the market is competitive, what is the price per gallon of milk and the number of gallons sold?

(b) With the connivance of some politicians, the diary farmers are able to form and maintain a cartel.(Such a cartel actually operates in the northeastern United States.) What is the cartel price, and how many gallons of milk are purchased?[Hint: The marginal revenue curve (MR) is given by MR=10-Q/5. Also, remember that the supply curve shows the marginal cost associated with each level of output]

(c) What are the rent associated with the cartel?

(d) Suppose that in order to maintain the cartel, the diary farmers simply give lump-sum campaign contributions to the relevant politicians. What is the maximum contribution they would be willing to make? What is the deadweight loss of the cartel?

4. (Rosen 6.10 改編)

Consider a society with 3 people (John, Eleanor, and Abigali) who use majority rule to decide how much money to spend on public park.

There are 3 options for spending on a public park: H(high), M(median), L(low).

These individuals rank the 3 options in the following way:

Rank	John	Eleanor	Abigali
1	M	L	H
2	L	H	M
3	H	M	L

Consider all possible pairwise elections: M vs. H, H vs. L, L vs. M. What is the outcome of each election? Does it appear, in the case, that majority rule would lead to a stable outcome on spending on the public park? If so, what is that choice? Would giving one person the ability to set the agenda affect the outcome? Explain

5. (Rosen 6.5)

In 2005, Kuwaiti women won the right to vote in parliamentary elections. Indeed, women voters now outnumber men voters in Kuwait because women are automatically registered while men have to register on their own. One woman noted, “The Ministers of Parliament used to vote against us; now they are wooing us to vote for them” [Fattah, 2006]. What does this tell us about the validity of the predictions of the median voter theorem?

6. (112-1期中考第4題)

Suppose there are five people A, B, C, D, and E in the society, and the following table shows their true values of a bridge construction plan:

people	A	B	C	D	E
v*($)	30	40	50	60	70

Suppose the cost of the bridge is $205.

Use what you learned from the Clarke-Groves mechanism to answer the following questions.

(a) Who are the decisive individuals?

(b) What are the payments of each individual?

(c) Is the total payment enough to cover the cost of the bridge?

(d) In general, what are the main problems of the Clarke-Groves mechanism?

7. 市場內有 X 財貨與 Y 財貨，且已知兩財貨的需求函數分別為$X(P_X, M)$與 $Y(P_Y, M)$，M 是所得，故可以知道預算限制式為$P_X \cdot X + P_Y \cdot Y = M$。

X 財貨份額為$(P_X \cdot X)/M$。請以數學推導以下兩命題:[提示:整理一階微分的結果]

(1) 當 X 財貨的價格彈性小於 1 時，X 財貨份額隨 X 財貨價格增加而增加

(2) 當 X 財貨的預算彈性大於 1 時，X 財貨份額隨所得增加而增加